

# Characterization of Charge-Coupled Device and the Verification of Gaussian Beam Propagation Relation

Wong Yin Lam<sup>a</sup>

<sup>a</sup>The Chinese University of Hong Kong,

## Abstract

Charge-coupled device(CCD) is widely used in ground-based and space telescopes. It act as an important scientific tool to unveil the mystery of the universe. However, some intrinsic defects may affect the performance of CCD, including but not limited to bad pixels, dark current and flat field effect. This project includes the measurement of the dark count, the linearity of the CCD sensor with increasing exposure time, and the calculation of the system gain from signal-variance plot. Moreover, the CCD is used as a tool to verify the Gaussian beam propagation relation of the He-Ne laser. The above results are obtained with satisfaction while the linearity and the Gaussian beam propagation relation is verified. Overall, the findings meet the theoretical expectations but there is still room for improvement in the experimental process.

**Keywords:** Charge-coupled devices, CCD, Gaussian beam propagation relation, Helium Neon laser, flat-field effect, dark current

## 1. Introduction

The charged-coupled devices(CCDs) are used in digital cameras, scanner and many other medical and industrial equipment. Particularly, CCDs revolutionized astronomical imaging and have led to the emergence of those astonishing astronomical images and discoveries. However, the performance and the sensitivity of the detector varies pixel by pixel and depends on wavelengths, so a lot of calibration and data processing is needed before the astronomical images are ready to use for scientific purposes.

In this project, several CCD intrinsic flaws, including the bias frame, the dark current and the flat field effect will be visited. Moreover, the performance of the CCD camera will be measured by its linearity till saturation. On the other hand, the system gain(the conversion between the electrons detected and the count recorded) of the camera will also be measured. Finally, the camera will be utilized as a scientific tool to help verify the Gaussian beam propagation relation of a Helium Neon laser.

## 2. Theory

CCD is a solid state imaging detector, making use of the photoelectric effect to measure the intensity of light shining on it. When the silicon layer is hitted by photons, electrons are liberated and stored in the potential well. After the exposure is done, the electrons held in each pixel will be shifted to the register row by row and the voltage produced by the electrons is measured and converted to count[2, 5].

Although CCD outcompeted photographic plates in astronomical imaging due to its accuracy, stability, sensitivity, broad spectral response, low power consumption and many other factors, CCD is not perfect. It has problems such as bad pixels,

thermal noise and inconsistent sensitivities among pixels. Scientists have to develop ways to calibrate the data and get rid of these defects, in which some of them will be examined in the next section (Section 3).

The conversion ratio between electrons and count of the CCD can be measured experimentally by obtaining the mean signal and the corresponding variance at different light intensity. Denote  $S_c$  and  $S_e$  as the signal measured in count and electron units respectively. Similarly, denote  $N_c$  and  $N_e$  as the total noise in count and electron units respectively. Signal and noise can be related in the following way

$$S_e = gS_c, \quad N_e = gN_c,$$

where  $g$  is the unknown gain of the CCD. The total noise can be ascribed to three sources: the readout noise  $R_e$ , the photon noise  $\sigma_e$  and the flat field noise  $\sigma_{O,e}$  such that

$$N_e^2 = R_e^2 + \sigma_e^2 + \sigma_{O,e}^2.$$

Meanwhile photon noise  $\sigma_e$  obeys the Poissonian statistics, it can be re-expressed by  $\sigma_e^2 = S_e$ , such that

$$N_e^2 = R_e^2 + S_e + \sigma_{O,e}^2.$$

Re-expressed the above equation in terms of gain and in count units, we obtain

$$N_c^2 = \frac{1}{g}S_c + R_c^2 + \sigma_{O,c}^2.$$

Therefore, for a coarse estimation, assuming the other two noise terms are independent of signal  $S$ , the gain can be obtained by the slope of a signal-variance plot.

However, the flat field noise  $\sigma_{O,c}$  is fixed and always proportional to the signal, therefore it can be expressed by  $\sigma_{O,c} = kS_c$

and the total noise becomes

$$N_c^2 = \frac{1}{g} S_c + R_c^2 + k^2 S_c^2.$$

Therefore, either the system gain need to be solved from the above equation, or the flat field effect need to be eliminated before the estimation. [3]

### 3. Experiment

There will be two main parts of this project, characterization of the CCD camera and the verification of the Gaussian beam propagation relation of a Helium Neon(He-Ne) laser. In the first part, it is further divided into three subsections, including measuring the dark count, the linearity and the system gain of the CCD. In the later part, CCD is used as a scientific tool to examine the performance of a He-Ne laser.

#### 3.1. Characterization of a digital CCD

##### 3.1.1. Dark count of the CCD

Intrinsically, there maybe some bad pixels in a piece of CCD detector. These bad pixels can be identified by taking a bias frame, which is an image with zero exposure with no light shining on it. All non-zero counts on the bias frame are recognized as the inherent defect of that CCD detector. Therefore, it is needed to obtain and subtract the bias frame from all images taken by the CCD detector. On the other hand, the CCD chip may be heated up when taking images. The thermal energy excite electrons to produce signal, which is called the dark current or the thermal noise. By taking a longer exposure when no light illuminated on the CCD, the signal due to the thermal noise can be obtained(by subtracting the bias frame from the image taken).

Therefore, in this part, the CCD is covered so no photon can hit the sensor(Fig. 1). Then, a 60s dark frame is taken at room temperature and a bias frame with 0.05ms exposure(the shortest exposure time it can achieve) is taken. A 100 x 100 pixels region of interest(ROI) is chosen such that the mean count value of pixels within the ROI in each frame is computed. The dark count is calculated by

$$\text{Dark count} = \frac{C_d - C_b}{60} \times \text{System gain}, \quad (1)$$

where  $C_d$  and  $C_b$  are the mean counts of dark frame and bias frame respectively, and by dividing the net dark count by 60s and multiplying it by the system gain(in electrons per count), the dark count in electrons per second is obtained.

##### 3.1.2. Linearity of the CCD

In this part, the cover of the CCD camera is taken off and a quartz lamp and a piece of flat white paper are used to provide a uniform light source(shown in Fig. 2). Flat field frames are taken with various exposure depth until the camera saturated.

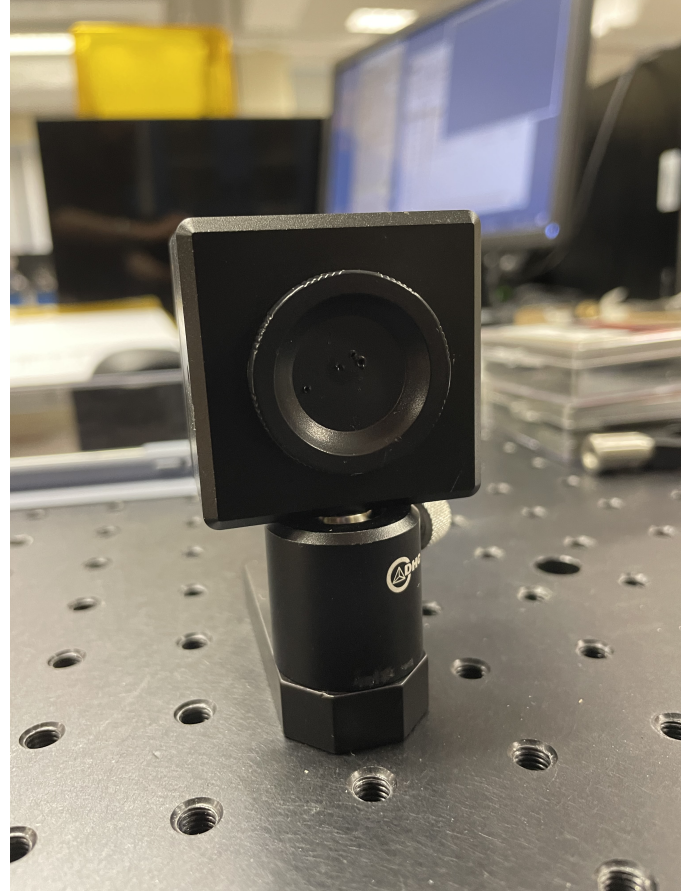


Figure 1: The CCD camera with the lid on.

##### 3.1.3. System gain of CCD

The gain of CCD is the conversion ratio of the photoelectrons to the read out count. As different CCD chips may have different gain, it is crucial to convert the count from a CCD camera to electrons before doing comparison. In the following, method 1 provides a crude estimation of the gain while method 2 are derived from a more rigorous examination of relation between signal and various sources of noise. The derivation is shown in section 2.

###### Method 1: Signal Variance Method.

The same set-up as in Fig. 2 is used. However, instead of one paper, nine papers are stacked over the CCD camera. Again, a bias frame is needed. Then, a frame is taken with 100ms exposure at each light intensity where the light intensity is altered by removing a piece of paper at a time. The gain is obtained by the slope of net signal against the variance.

However, by this method, the flat field effect hasn't been taken into account, so the gain obtained will be underestimated. Therefore, different approaches are used to address the issue where one of them are elucidated below.

###### Method 2: Correction of the flat field effect.

In this method, flat field effect is overcome by subtracting two images obtained in succession at each signal level. For

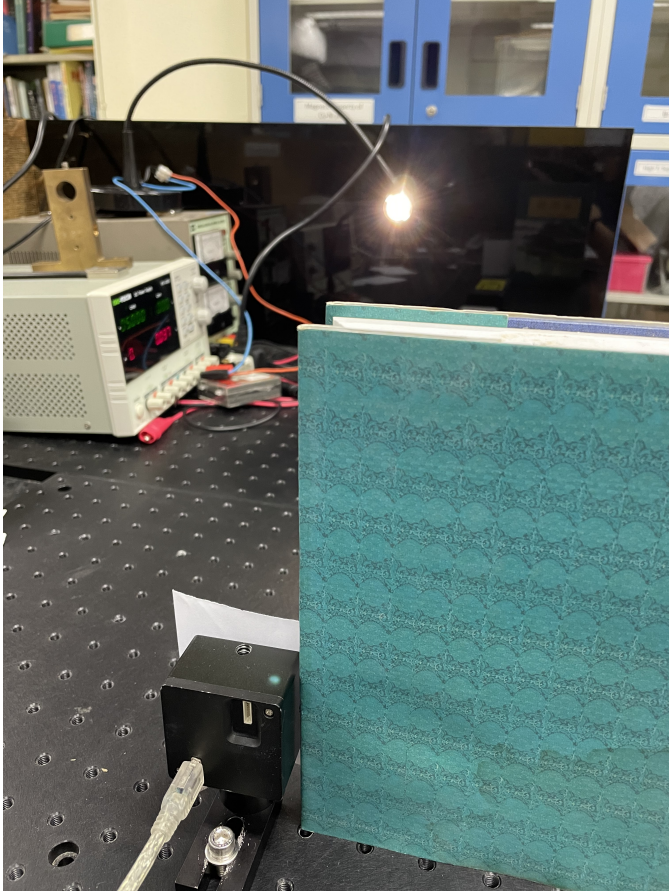


Figure 2: The set-up in Sect. 3.1.2.

each image, the mean signal level in the ROI is obtained as  $S_A$  and  $S_B$ . Then, compute the ratio between the two signal levels by

$$r = \frac{S_A}{S_B}$$

and multiply the image B by  $r$ , which corrects image B to the same signal level as image A without affecting its noise structure. Image B is then subtracted from image A to cancel out the flat field effect. After the above procedures, the rest follows the steps in method 1.

### 3.2. Verification of the Gaussian beam propagation relation

A 632nm He-Ne laser is used in this part. As laser beam is a strong light source, it will easily saturate the CCD camera. Therefore, neutral density filters(ND filters) are needed to attenuate the strong laser beam. In this investigation, the intensity of laser beam is reduced to 0.01% of its original. Images of the laser beam are taken at different distances and the beam radius is measured.

For the Gaussian propagation relation, the beam radius  $\omega$  is a function of  $z$ , where  $z_0(z=0)$  is defined at where  $\omega = \omega_0$ , i.e. the minimum beam radius or the beam waist[1, 4]

$$\omega(z) = \omega_0 \sqrt{1 + \left( \frac{M^2 z \lambda}{\pi \omega_0^2} \right)^2}, \quad (2)$$

where  $\lambda$  is the wavelength of the laser and  $M^2$  is a parameter quantifying the beam quality.

However, in reality, the location where  $\omega = \omega_0$  is unknown before the measurement. Therefore, the distance measured in this experiment is not the  $z$  in Eq. 2. Consequently, the equation used in this experiment is actually

$$\omega(\tilde{z}) = \omega_0 \sqrt{1 + \left( \frac{M^2 \tilde{z} \lambda}{\pi \omega_0^2} \right)^2}, \quad (3)$$

where  $\tilde{z} = z - z_0$  and  $\tilde{z}$  is measured from the laser aperture.

## 4. Results and discussions

### 4.1. Characterization of a digital CCD

#### 4.1.1. Dark count of the CCD

By taking the mean count value for bias frame and the bias-subtracted dark frame,  $C_b = 0.0014$  and  $C_d = 4.4593$ . By applying Eq. 1, the dark count calculated is 2.535 electrons per second. Note that the system gain in Method 2 of section 3.1.3, which is 34.113 electrons per count, is used here.

#### 4.1.2. Linearity of the CCD

The following plot shows how the mean count values grow with increasing time. From the plot, it can be observed that a good linearity is reached by the CCD camera. It is worth noting that the  $R^2 = 1.000$  indicates an excellent performance in the consistency of the CCD camera over different exposure depth.

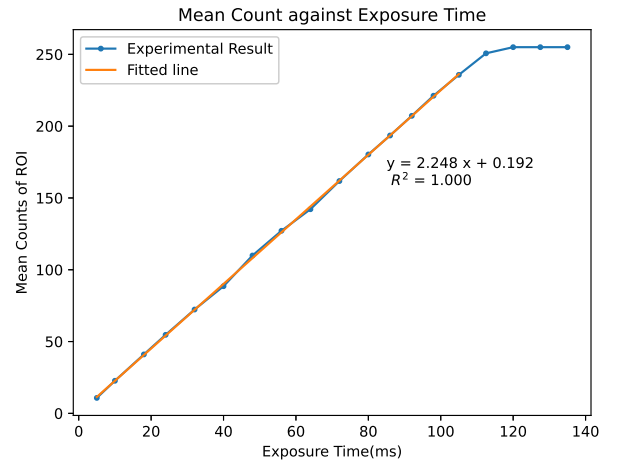


Figure 3: Plot of mean count of the region of interest(ROI) against the exposure time(ms).

However, in Fig. 2, the quartz lamp shined directly towards the CCD camera, which may cause a non-uniform illumination. Therefore, the position of the lamp in this experiment is not optimized.

#### 4.1.3. System gain of CCD

##### Method 1: Signal Variance Method.

The slope of the signal variance plot(Fig. 5) is the system gain of the CCD camera. Data in this part is tabulated in Appendix A. The gain obtained is  $29.816 \pm 0.232$  electrons per count.  $R^2 = 1.000$  indicates the fit is highly satisfactory. However, in this method, flat field noise is neglected so the slope underestimated the *real* system gain.

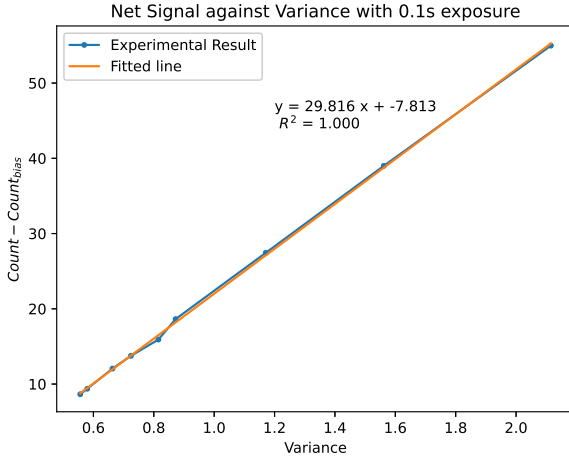


Figure 4: Plot of net mean count( $C - C_{bias}$ ) of ROI with 0.1s exposure against the variance over the ROI.

##### Method 2: Correction of the flat field effect.

In this method, the flat field effect is taken into account so the gain obtained should resembles the real gain to a larger extent. Data in this part is also tabulated in Appendix A. The gain obtained in this method is  $34.113 \pm 1.219$  electrons per count. Although the  $R^2$  value is a bit lower than the last fit,  $R^2 = 0.991$  still indicates a satisfactory fit.

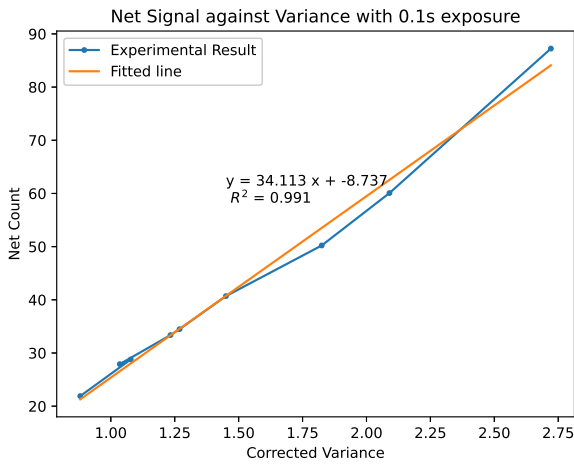


Figure 5: Plot of net count of ROI with 0.1s exposure against the corrected variance over the ROI.

Comparing the gain value obtained in two methods, they are inconsistent even within the uncertainties. Therefore, the effect of the flat field noise cannot be neglected when obtaining the system gain of the CCD sensor. Moreover, as dark currents are not subtracted in these two investigations, it adds to both the signal and variance, therefore the slope obtained may not reflect the real gain.

#### 4.2. Verification of the Gaussian beam propagation relation

The laser beam is captured using the CCD camera, just as shown in Fig. 6. Then, a Gaussian function is fitted to each image and the beam diameter is measured as  $[x_0 - 3\sigma, x_0 + 3\sigma]$ (i.e. 99.7% of light is included)(see Fig. 7). All fitted graphs are presented in Appendix C. The beam diameter in pixel unit is converted to that in physical unit by multiplying the pixel width  $3.75 \mu m$ . Then, the data points are fitted with Eq. 3.

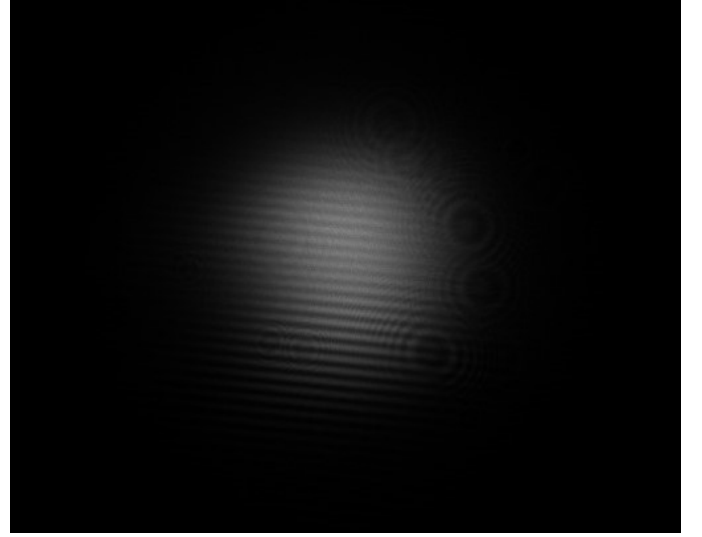


Figure 6: The laser beam captured by the CCD, image taken at 34cm from the laser aperture.

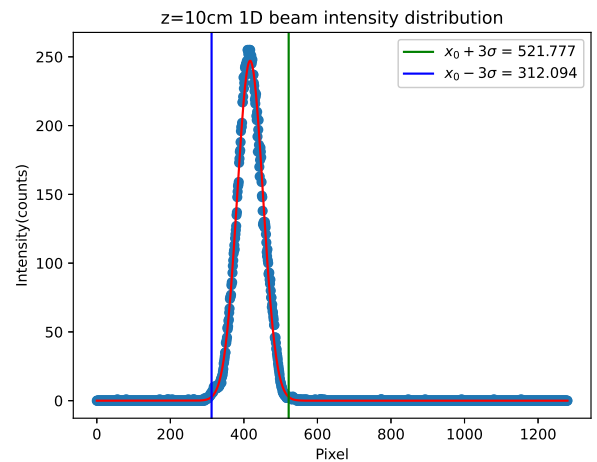


Figure 7: Gaussian fit of the laser beam captured by the CCD, image taken at 10cm from the laser aperture.



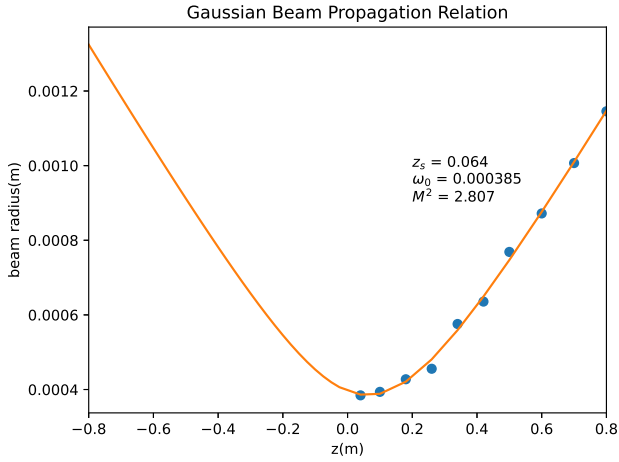


Figure 8: A plot of beam radius against distance to the laser aperture fitted with eq. 3.

From the fit, the minimum beam radius  $\omega_0$  obtained is 0.385 mm, which has the same order as the theoretical value 0.24 mm. For the  $M^2$  factor, which quantify the beam quality(1 is perfect), is 2.807, which is still acceptable. Furthermore, the real  $z_0$  are located at 6.4cm in front of the laser aperture. However, the percentage error of the  $\omega_0$  obtained is

$$\Delta = \frac{|r_{exp} - r_{theo}|}{r_{theo}} \times 100\% = 60.4\%$$

which is reasonably large. As measurement cannot be taken inside the glass tube inside the laser, therefore by fitting the function with only half the data, the accuracy is limited. Still, the Gaussian beam propagation relation of the He-Ne laser beam is verified.

## 5. Conclusion

The dark count obtained is 2.535 electrons per second by using the 34.113 electrons per count(the result obtained in Method 2 in section 3.1.3) as the system gain. On the other hand, the linearity of the CCD camera is reached with high satisfaction. For obtaining the system gain, two methods are adopted, one gives 29.816 electrons per count while another gives 34.113 electrons per count. Both fits are beyond satisfactory. Nevertheless, the later one is more valued as the former one unable to account for the flat field noise, which leads to an underestimation of the slope of the signal variance plot, i.e. the system gain. For the He-Ne laser beam, Gaussian fitting is used to obtain the beam radius and the Gaussian beam propagation relation between the beam radius and the distance from the laser is verified. Generally, most of the objectives in this project are met and the results agree with the expectations.

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### Appendix A. Data in each part

|                     |         |         |         |         |         |         |         |        |        |
|---------------------|---------|---------|---------|---------|---------|---------|---------|--------|--------|
| Variance            | 2.115   | 1.562   | 1.171   | 0.872   | 0.815   | 0.724   | 0.663   | 0.580  | 0.556  |
| Mean Signal(counts) | 54.9715 | 38.9968 | 27.4405 | 18.6348 | 15.9115 | 13.7524 | 12.0518 | 9.3757 | 8.6484 |

Table A.1: Signals and variances in **Method 1** of section 3.1.3

|                     |         |         |         |         |         |         |         |         |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Variance            | 2.721   | 2.090   | 1.825   | 1.450   | 1.269   | 1.233   | 1.035   | 1.078   | 0.880   |
| Mean Signal(counts) | 87.2402 | 60.0742 | 50.2211 | 40.6973 | 34.4829 | 33.3662 | 27.9083 | 28.7889 | 21.9031 |

Table A.2: Signals and variances in **Method 2** of section 3.1.3

|                            |        |       |       |        |         |         |        |         |         |         |
|----------------------------|--------|-------|-------|--------|---------|---------|--------|---------|---------|---------|
| $\tilde{z}(\text{in cm})$  | 4      | 10    | 18    | 26     | 34      | 42      | 50     | 60      | 70      | 80      |
| beam diameter(in $\mu m$ ) | 768.75 | 787.5 | 855.0 | 911.25 | 1151.25 | 1271.25 | 1537.5 | 1743.75 | 2013.75 | 2291.25 |

Table A.3: Laser beam diameter obtained by gaussian fitting

### Appendix B. Codes used for the project

The following github link <https://github.com/marteswong/CCD> includes all the codes used in the project, including but not limited to conversion between hexadecimal to decimal, computing the mean and variances, do curve fitting and outputing the relevant plots.

### Appendix C. Gaussian Fitted Beam Profile

